

Mahad Rana

551-500-8605 | mahadmrana@gmail.com | [linkedin.com/in/mahad-rana](https://www.linkedin.com/in/mahad-rana) | github.com/MahadRana

EDUCATION

Stevens Institute of Technology

Hoboken, NJ

Bachelor of Science in Computer Science

Expected: May 2026

- GPA: 4.0
- Honors: Dean's List, Edwin A. Stevens Scholarship, Presidential Scholarship, Pinnacle Scholars
- Relevant Courses: Data Structures, Algorithms, Discrete Structures, Deep Learning, Systems Programming

EXPERIENCE

Software Development Intern

May 2024 - August 2024

Croll-Reynolds

Parsippany-Troy Hills, NJ

- Increased application productivity by 25% by expanding a .NET program to handle 4-stage vacuum computations and automatically generate technical Excel datasheets.
- Boosted program security by implementing time-based locks and a serial number security layer, safeguarding sensitive data and preventing unauthorized access.
- Improved satisfaction by reducing the number of user-reported bugs by 30% through the resolution of issues in calculations for process vacuum system components within the application.

Undergraduate Machine Learning Researcher

May 2023 - August 2023

Stevens Institute of Technology

Hoboken, NJ

- Formulated image processing algorithms using the OpenCV library in Python to preprocess a dataset of 100+ high-resolution images, resulting in reduced noise and improved model performance.
- Implemented an image segmentation machine learning model using the PyTorch library in Python that can create binary contours from images of guidewires with an accuracy rate of 95%.
- Developed an image-processing algorithm using MATLAB that creates a smooth midline from a contour of a guidewire for further calculations.

PROJECTS

Full-Stack Prayer Tracker | *JavaScript/React, Node.js, Express.js, MySQL, HTML/CSS, Docker*

- Created a responsive frontend for a prayer tracking application using React, implementing advanced state management with Context API and custom hooks to provide a smooth user experience.
- Configured CRUD operations in RESTful API endpoints using Express.js to ensure seamless communication between the MySQL database and the user interface, providing accurate and timely updates.
- Utilized Docker to create isolated environments for the frontend, backend, and database services, streamlining development, deployment, and scalability.

QuizConnect | *C, Linux, Networking (TCP/IP)*

- Built a client-server trivia game using socket programming in C, establishing reliable TCP/IP connections to facilitate real-time communication between clients and the server.
- Engineered a communication protocol enabling the server to distribute trivia questions to clients and receive responses efficiently, ensuring smooth data transmission and handling potential errors.
- Implemented multi-threading on the server side to manage multiple clients simultaneously, maintaining consistent game logic and synchronization across all connected players.

HelpDesk AI | *AWS, JavaScript/React, Node.js, Express.js, HTML/CSS*

- Deployed and maintained a full-stack application on AWS EC2, ensuring secure communication between services by configuring security groups to control inbound and outbound traffic, and using PM2 for process management.
- Integrated GPT models with an Express.js backend using OpenAI's API, enabling intelligent responses for troubleshooting queries with a focus on secure and efficient request handling.
- Engineered a React frontend with real-time chat functionality and advanced state management, improving responsiveness and enabling efficient, interactive communication.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, HTML/CSS, JavaScript, MATLAB, R, Ocaml, Bash, VB.NET, XML, C#, SQL

Frameworks: React, Express.js, JUnit, TensorFlow, PyTorch, OpenCV, Pandas, NumPy, Matplotlib, WPF, .NET

Tools: Git, AWS, MongoDB, Node.js, Linux, Ubuntu, Unix, Visual Studio, Docker